

# IT AUDIT CASE STUDY 1

## I. Project Title

Internal Audit of Information Security Management System at TechFiction Corp

## 2. Background

TechFiction Corp is a mid-sized IT services company managing:

- Customer data / PII (Personally Identifiable Information).
- Cloud infrastructure.
- Internal HR systems.
- Financial systems.

Recent issues:

- Incidents of unauthorized access.
- Missing Audit logs.
- Delayed Patching.
- Vulnerable / Weak password policies.

Audit Committee has requested an internal ISMS audit to assess:

- Effectiveness of risk management.
- Internal controls.
- Compliance with *ISO/IEC 27001* principles.

## 3. Audit Objectives

- a. Evaluate information security controls effectiveness.
- b. Assess compliance with policies.
- c. Identify security risks and vulnerabilities.
- d. Recommend corrective actions.

## 4. Audit Scope

Included:

- User Access Management.
- Data Security.
- Network Security.
- Incident Management.

Excluded:

Physical security (limited resources).

5. Dataset

- Dataset 1: User Access Logs.
- Dataset 2: System Vulnerabilities.
- Dataset 3: Incident Records.

6. Standard Audit Methodology

Step 1: Planning

- Define objectives.
- Identify risks.
- Prepare audit checklist.

Step 2: Fieldwork

- Data analysis.
- Sampling.
- Control testing.

Step 3: Reporting

- Findings.
- Risk rating.
- Recommendations.

Step 4: Monitoring

- Follow-up actions.

7. RACM (Risk and Control Matrix)

| Risk                   | Control                                | Test                  |
|------------------------|----------------------------------------|-----------------------|
| Unauthorized access    | MFA (Multi-factor Authentication)      | Check MFA status      |
| Data breach            | Firewall                               | Verify firewall       |
| Delayed patching       | Patch policy                           | Check patch period    |
| Incident mismanagement | Response SLA (Service Level Agreement) | Analyze response time |

## 8. Analysis

- a. Identify High-Risk Users.
- b. Calculate Deviation Rate:
  - Apply random sampling.
  - Confidence level (95%).

$$Deviation = \frac{Errors}{Total\ samples} * 100\%$$

- c. Check Vulnerability Risk Score.
- d. Analyze incident risk.
- e. Visualize.

## 9. Deliverables

You should produce:

- a. .ipynb (Jupyter notebook file).
- b. .pdf.
- c. .pptx (Presentation file).

## 10. How to work (Step-by-step)

- a. Run analysis code.
- b. Interpret results.
- c. Write findings (5 C's format).
- d. Create final report.

## 11. Final Report Structure

Sections:

1. Cover Page.
2. Executive Summary.
3. Scope & Objectives.
4. Methodology.
5. Findings.
6. Recommendations.
7. Conclusion.